Data Communication & Computer Networks — Topic 7: Network Security

Practice Test Booklet — 146 MCQs (2-column, without solutions) · Easy + Medium + Hard mix · Har question ke saath **Source** · Print ke liye · Reference: Forouzan + India-level & worldwide exam papers (GATE/NET/KVS/EMRS/DSSSB/SSC/RRB/IBPS) · Prepared for: RPSC Programmer • ACP • RSMSSB IA • SA • DD • BCI • SCI • REET • RAS • DSSSB • KVS • NVS • HTET • SSC CGL/CHSL • RRB • IBPS • UPSC & all Rajasthan + Central Govt. Exams (Computer Awareness)

Section A — Security Goals & Attacks

Q1. In the virus life cycle, which phase actually performs the malicious activity?

Virus life cycle में कौन-सा phase वास्तव में malicious activity करता है?

(A) Infection Phase **(B)** Triggering Phase
(C) Dormant Phase **(D)** Execution (Payload) Phase

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

Q2. Which backup medium provides Write-Once-Read-Many (WORM) functionality?

कौन-सा backup medium Write-Once-Read-Many (WORM) functionality देता है?

(A) Hard disk **(B)** Magnetic tape
(C) Optical disk (CD-R / DVD-R) **(D)** Flash drive

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

Q3. Which attack becomes possible if the SAME key is reused many times in symmetric encryption?

यदि symmetric encryption में एक ही key कई बार उपयोग हो, तो कौन-सा attack संभव हो जाता है?

(A) Replay attack **(B)** Known-plaintext attack
(C) Man-in-the-middle attack **(D)** Phishing attack

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

Q4. Which of the following tools is mainly used for password cracking?

निम्न में से कौन-सा tool मुख्यतः password cracking के लिए उपयोग होता है?

(A) Wireshark **(B)** Nmap
(C) Cain and Abel **(D)** Angry IP Scanner

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

Q5. Which port and IP address scanner is most famous among users?

Users के बीच सबसे प्रसिद्ध port और IP address scanner कौन-सा है?

(A) Cain and Abel **(B)** Angry IP Scanner
(C) Snort **(D)** Ettercap

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

Q6. The term 'CIA triad' in information security refers to:

Information security में 'CIA triad' शब्द किसे कहते हैं?

(A) Cryptography, Integrity, Authentication **(B)** Confidentiality, Integrity, Availability
(C) Confidentiality, Information, Access **(D)** Control, Information, Access

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

Q7. A modification attack violates which security goal?

Modification attack कौन-से security लक्ष्य का उल्लंघन करता है?

(A) Confidentiality **(B)** Integrity
(C) Availability **(D)** Authentication

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

Q8. Which of the following attacks violates confidentiality?

निम्न में से कौन-सा attack confidentiality का उल्लंघन करता है?

(A) Interception **(B)** Modification
(C) Fabrication **(D)** Interruption

[Source: Subject Test Solution-1 Part-2 (CN & Security) slides] · Medium

Q9. A Denial of Service (DoS) attack violates which goal?

Denial of Service (DoS) attack कौन-से लक्ष्य का उल्लंघन करता है?

(A) Confidentiality **(B)** Integrity
(C) Availability **(D)** Authentication

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

Q10. A spoofing attack targets which security aspect?

Spoofing attack किस security पहलू को लक्षित करता है?

(A) Authentication **(B)** Integrity
(C) Availability **(D)** Confidentiality

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

Q11. Which of the following is an ACTIVE attack?

निम्न में से कौन-सा ACTIVE attack है?

(A) Interception **(B)** Modification
(C) Traffic analysis **(D)** Eavesdropping

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

Q12. Match the attacks with their descriptions: (a) Phishing (b) DoS Attack (c) Man-in-the-Middle (d) Spoofing — (i) Flooding network traffic (ii) Stealing user credentials (iii) Intercepting communication (iv) Impersonation attack

हमलों को उनके विवरण से मिलाइए: (a) Phishing (b) DoS Attack (c) Man-in-the-Middle (d) Spoofing — (i) नेटवर्क ट्रैफ़िक भरना (ii) यूज़र जानकारी चुराना (iii) संचार में हस्तक्षेप (iv) पहचान की नकल

(A) a–ii, b–i, c–iii, d–iv **(B)** a–i, b–ii, c–iii, d–iv
(C) a–iii, b–i, c–ii, d–iv **(D)** a–ii, b–iii, c–i, d–iv

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch) — Q3 match (codes recovered)] · Medium

Q13. 'Junk email' is also known as:

'Junk email' को और क्या कहते हैं?

(A) Spam **(B)** Spoof
(C) Sniffer **(D)** Spool

[Source: RRB NTPC computer banks (via practicemock PYQ set)] · Medium

Q14. A program that replicates itself across networks without needing a host file is a:

वह program जो बिना host file के network पर खुद की प्रतियाँ बनाता है:

(A) Virus **(B)** Worm
(C) Trojan **(D)** Cookie

[Source: Practice — original (textbook-based)] · Medium

Q15. Eavesdropping on network traffic is which type of attack?

Network traffic की जासूसी कौन-सा attack है?

(A) Passive attack **(B)** Active attack
(C) DoS attack **(D)** Replay attack

[Source: Practice — original (textbook-based)] · Medium

Q16. Sending a fake e-mail that appears to come from a bank to steal passwords is called:

बैंक के नाम से नक़ली e-mail भेजकर password चुराना कहलाता है:

(A) Phishing **(B)** Sniffing
(C) Spoofing **(D)** Tunneling

[Source: Practice — original (textbook-based)] · Medium

Q17. Flooding a server with so many requests that legitimate users cannot use it is a:

Server को इतने requests से भर देना कि असली users न चला सकें:

(A) DoS attack **(B)** Phishing
(C) Sniffing **(D)** Encryption

[Source: Practice — original (textbook-based)] · Medium

Q18. A SYN flood (denial-of-service) attack exploits which part of network communication?

SYN flood (denial-of-service) attack संचार के किस भाग का दुरुपयोग करता है?

(A) The TCP three-way handshake / TCP three-way handshake **(B)** The DNS resolution process / DNS resolution process
(C) The ARP cache / ARP cache **(D)** The ICMP echo request / ICMP echo request

[Source: Practice — original (textbook-based)] · Medium

Q19. A honeypot in network security is:

Network security में honeypot होता है:

(A) A decoy system set up to attract attackers and study their activity / एक प्रलोभन (decoy) system जो attackers को आकर्षित कर उनकी गतिविधि अध्ययन करता है **(B)** A firewall rule that blocks all traffic / सभी traffic रोकने वाला firewall rule
(C) A backup server / एक backup server **(D)** An antivirus quarantine folder / Antivirus quarantine folder

[Source: Practice — original (textbook-based)] · Medium

Q20. A shadow honeypot combines a honeypot with:

Shadow honeypot जोड़ता है honeypot के साथ:

(A) An anomaly detection system (suspect traffic is run inside the honeypot) / Anomaly detection system (संदिग्ध traffic को honeypot के अंदर चलाया जाता है) **(B)** A physical lock / एक भौतिक lock
(C) A DNS sinkhole only / सिर्फ DNS sinkhole **(D)** A load balancer / एक load balancer

[Source: Practice — original (textbook-based)] · Medium

Q21. The goal of 'Integrity' ensures that:

'Integrity' का लक्ष्य सुनिश्चित करता है कि:

(A) Data can be accessed only by authorised users
(B) Data is accurate and has not been modified
(C) Data is always available when needed **(D)** Unauthorised users cannot enter the system

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Hard

Q22. Which of the following is NOT a goal of network security?

निम्न में से कौन-सा network security का लक्ष्य नहीं है?

(A) Confidentiality **(B)** Integrity
(C) Availability **(D)** Redundancy

[Source: Subject Test Solution-1 Part-2 (CN & Security) slides] · Hard

Q23. The primary difference between active and passive attacks is:

Active और passive attacks के बीच मुख्य अंतर क्या है?

(A) Active attacks change data; passive attacks do not
(B) Passive attacks destroy hardware
(C) Both are the same **(D)** Passive attacks are legal

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Hard

Q24. A _____ does not perform any destructive action directly; it only consumes system resources to bring the system down.

एक _____ सीधे कोई विनाशकारी काम नहीं करता; वह सिस्टम को गिराने के लिए सिर्फ system resources खाता है.

(A) Virus **(B)** Worm
(C) Trojan **(D)** Spyware

[Source: HTET 2024 (via KnowledgeGate solved PYQ bank)] · Hard

Q25. Ensuring that data has not been altered in transit relates to which security goal?

यह सुनिश्चित करना कि data रास्ते में बदला नहीं, किस security goal से है?

(A) Confidentiality **(B)** Integrity
(C) Availability **(D)** Authentication

[Source: Practice — original (textbook-based)] · Hard

Section B — Cryptography

Q26. The text that gets transformed using a cipher algorithm is called:

Cipher algorithm से जिस text को बदला जाता है उसे क्या कहते हैं?

(A) Complex Text **(B)** Transformed Text
(C) Plain Text **(D)** Scalar Text

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch) — Q9] · Easy

Q27. In the digital signature technique, when the whole message is signed using an asymmetric key, the receiver uses:

Digital signature technique में, जब पूरा संदेश asymmetric key से sign होता है, तो प्राप्तकर्ता उपयोग करता है:

(A) Her or his own public key **(B)** Her or his own private key
(C) The sender's public key **(D)** None of these

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

Q28. Which statement about plaintext, ciphertext and key is MOST accurate?

Plaintext, ciphertext और key के बारे में कौन-सा कथन सबसे सही है?

(A) Plaintext is generated after decryption using a public key **(B)** Ciphertext is the original readable message
(C) Plaintext is transformed into ciphertext using an algorithm and a key **(D)** Ciphertext and plaintext are always of the same length

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch) — Q10] · Medium

Q29. Which of the following uses the SAME key for encryption and decryption?

निम्न में से कौन-सा encryption और decryption के लिए एक ही key उपयोग करता है?

(A) Symmetric key cipher **(B)** Asymmetric key cipher
(C) Hash function **(D)** Digital signature

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

Q30. Which of the following uses a PAIR of keys (public and private)?

निम्न में से कौन-सा keys की एक जोड़ी (public और private) उपयोग करता है?

(A) Symmetric cipher **(B)** Asymmetric cipher
(C) DES **(D)** AES

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

Q31. Which is FASTER in encryption and decryption?

Encryption और decryption में कौन-सा तेज़ है?

(A) Symmetric key cipher **(B)** Asymmetric key cipher
(C) Cannot decide **(D)** None of these

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

Q32. What is the main disadvantage of symmetric key encryption?

Symmetric key encryption की मुख्य कमी क्या है?

(A) It is very slow **(B)** Key distribution is difficult
(C) It uses two keys **(D)** It cannot encrypt text

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

Q33. In public-key cryptography, a message encrypted with the public key can only be decrypted by:

Public-key cryptography में public key से encrypt किया संदेश केवल किसके द्वारा decrypt हो सकता है?

(A) Another public key **(B)** The same public key
(C) The private key corresponding to that public key
(D) A session key

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

Q34. Data Encryption Standard (DES) is a:

Data Encryption Standard (DES) क्या है?

(A) Block cipher **(B)** Stream cipher
(C) Bit cipher **(D)** Byte cipher

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

Q35. Which cipher method encrypts one bit or byte at a time?

कौन-सी cipher method एक बार में एक bit या byte encrypt करती है?

(A) Block cipher **(B)** Stream cipher
(C) Substitution cipher **(D)** Transposition cipher

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

Q36. The Caesar Cipher is an example of which cryptographic technique?

Caesar Cipher किस cryptographic technique का उदाहरण है?

(A) Transposition cipher **(B)** Substitution cipher
(C) Stream cipher **(D)** Block cipher

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

Q37. The process of detecting hidden data inside a file is known as:

किसी file के भीतर छिपे data का पता लगाने की प्रक्रिया को क्या कहते हैं?

(A) Cryptanalysis **(B)** Steganalysis
(C) Data mining **(D)** Steganography

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

Q38. What is the output of a steganography process called?

Steganography प्रक्रिया के output को क्या कहते हैं?

(A) Ciphertext **(B)** Stego-object
(C) Encoded file **(D)** Key file

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

Q39. In steganography, the file used to hide the secret data is called:

Steganography में secret data छिपाने के लिए जिस file का उपयोग होता है उसे क्या कहते हैं?

(A) Secret file **(B)** Cover file
(C) Cipher file **(D)** Key file

[Source: Subject Test Solution-1 Part-2 (CN & Security) slides] · Medium

Q40. Which of the following is an example of asymmetric encryption?

निम्न में से कौन-सा asymmetric encryption का उदाहरण है?

(A) AES **(B)** DES
(C) RSA **(D)** Blowfish

[Source: SSC CGL Tier-2 repeated Qs (via Oliveboard) · also BCI_SCI ST-01] · Medium

Q41. The data to be encrypted at the sender's site is called _____ and the data to be decrypted at the receiver's end is called _____.

Sender की ओर encrypt किए जाने वाले data को _____ और receiver की ओर decrypt किए जाने वाले data को _____ कहते हैं.

(A) Secret key, public key **(B)** Public key, secret key
(C) Plain text, cipher text **(D)** Cipher text, plain text

[Source: SBI PO Computer Awareness PYQ (via bankersadda)] · Medium

Q42. Which algorithm is symmetric?

निम्न में से कौन symmetric algorithm है?

(A) RSA **(B)** AES
(C) Diffie-Hellman **(D)** ECC

[Source: Practice — original (textbook-based)] · Medium

Q43. A hash function is mainly used to provide:

Hash function मुख्यतः देता है:

(A) Confidentiality **(B)** Integrity
(C) Availability **(D)** Compression

[Source: Practice — original (textbook-based)] · Medium

Q44. A digital signature provides:

Digital signature देता है:

(A) Confidentiality only / केवल confidentiality **(B)** Authentication and non-repudiation / Authentication और non-repudiation
(C) Availability **(D)** Compression

[Source: Practice — original (textbook-based)] · Medium

Q45. DES is an example of which type of cipher?

DES किस प्रकार का cipher है?

(A) Stream cipher **(B)** Block cipher
(C) Asymmetric cipher **(D)** Hash function

[Source: Practice — original (textbook-based)] · Medium

Q46. Public key is used to:

Public key का उपयोग किसके लिए होता है?

(A) Encrypt the data **(B)** Decrypt the data
(C) Both encrypt and decrypt **(D)** Generate session keys only

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Hard

Q47. Which statement about symmetric and asymmetric cryptography is TRUE?

Symmetric और asymmetric cryptography के बारे में कौन-सा कथन सही है?

(A) Symmetric cryptography provides authentication
(B) Asymmetric cryptography is faster than symmetric
(C) Symmetric cryptography requires secure key exchange **(D)** Both use two different keys

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security) — Q33] · Hard

Q48. What is the key length of AES-256 encryption?

AES-256 encryption की key length कितनी है?

(A) 128 bits **(B)** 192 bits
(C) 256 bits **(D)** 512 bits

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Hard

Q49. Which of the following statements is FALSE?

निम्न में से कौन-सा कथन गलत है?

(A) AES is a block cipher **(B)** RSA is a symmetric cipher
(C) Diffie-Hellman is used for key exchange **(D)** DES is symmetric

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Hard

Q50. Suppose everyone in a group of N people wants to communicate secretly with the N−1 others using a symmetric-key cryptographic system. The number of keys required in the system as a whole is:

मान लीजिए N लोगों के समूह में हर व्यक्ति बाकी N−1 लोगों से symmetric-key cryptographic system से गुप्त बातचीत करना चाहता है. कुल कितने keys चाहिए होंगी?

(A) 2N **(B)** N(N − 1)
(C) N(N − 1)/2 **(D)** (N − 1)²

[Source: GATE CS PYQ (via examprev)] · Hard

Q51. The AES symmetric encryption standard supports key sizes of:

AES symmetric encryption standard की key sizes होती हैं:

(A) 128, 192 and 256 bits / 128, 192 और 256 bits **(B)** 56, 128 and 256 bits / 56, 128 और 256 bits
(C) 1024, 2048 and 4096 bits / 1024, 2048 और 4096 bits
(D) 64, 128 and 192 bits / 64, 128 और 192 bits

[Source: Practice — original (textbook-based)] · Hard

Section C — Firewalls, IPSec & Security Protocols

Q52. IPSec is MOST commonly used to implement:

IPSec मुख्यतः किसके लिए उपयोग होता है?

(A) Secure email **(B)** Secure web browsing
(C) Virtual Private Network (VPN) **(D)** Wireless security

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

Q53. In IPSec Tunnel mode:

IPSec Tunnel mode में:

(A) Only the payload is encrypted **(B)** Only the header is encrypted
(C) The entire IP packet is encrypted **(D)** No encryption is used

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

Q54. The SSL/TLS handshake protocol is mainly used for:

SSL/TLS handshake protocol मुख्यतः किसके लिए उपयोग होता है?

(A) Data encryption **(B)** Data compression
(C) Key exchange and authentication **(D)** Packet routing

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

Q55. Match the security mechanisms with their OSI layers: (a) Firewall (b) Digital Signature (c) MAC Filtering (d) Port-based Filtering — (i) Network Layer (ii) Application Layer (iii) Data Link Layer (iv) Transport Layer

Security mechanisms को OSI layers से मिलाइए: (a) Firewall (b) Digital Signature (c) MAC Filtering (d) Port-based Filtering — (i) Network Layer (ii) Application Layer (iii) Data Link Layer (iv) Transport Layer

(A) a–i, b–ii, c–iii, d–iv **(B)** a–ii, b–i, c–iii, d–iv
(C) a–iv, b–ii, c–i, d–iii **(D)** a–i, b–iii, c–ii, d–iv

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch) — Q4 match (codes recovered)] · Medium

Q56. Match the security protocols with their OSI layers: (a) IPSec (b) SSL/TLS (c) PGP (d) WPA2 — (i) Application Layer (ii) Network Layer (iii) Transport/Presentation Layer (iv) Data Link Layer (Wi-Fi)

Security protocols को OSI layers से मिलाइए: (a) IPSec (b) SSL/TLS (c) PGP (d) WPA2 — (i) Application Layer (ii) Network Layer (iii) Transport/Presentation Layer (iv) Data Link Layer (Wi-Fi)

(A) a–ii, b–iii, c–i, d–iv **(B)** a–ii, b–i, c–iii, d–iv
(C) a–iii, b–ii, c–i, d–iv **(D)** a–i, b–ii, c–iii, d–iv

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch) — Q1 match (codes recovered)] · Medium

Q57. IPSec operates in which mode(s)?

IPSec किस/किन mode(s) में काम करता है?

(A) Transport mode only **(B)** Tunnel mode only
(C) Both transport mode and tunnel mode **(D)** Application mode

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Medium

Q58. Which mode of IPSec provides end-to-end security between two hosts?

IPSec का कौन-सा mode दो hosts के बीच end-to-end security देता है?

(A) Transport mode **(B)** Tunnel mode
(C) Static mode **(D)** Gateway mode

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)] · Medium

Q59. Which of the following is used in HTTPS for encryption?

HTTPS में encryption के लिए निम्न में से क्या उपयोग होता है?

(A) SSL/TLS **(B)** FTP
(C) ARP **(D)** NAT

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)] · Medium

Q60. In which layer of the network architecture is the Secure Socket Layer (SSL) used?

Secure Socket Layer (SSL) network architecture की किस layer में उपयोग होती है?

(A) Physical layer **(B)** Session layer
(C) Application layer **(D)** Presentation layer

[Source: Unit Test-4 Solution (OCR), p6 (bank key)] · Medium

Q61. When a logic bomb is activated by a time-related event, it is known as a:

जब कोई logic bomb किसी समय-संबंधी घटना से सक्रिय होता है, तो उसे कहते हैं:

(A) Time-related bomb sequence **(B)** Virus
(C) Time bomb **(D)** Trojan horse

[Source: SBI PO Computer Awareness PYQ (via bankersadda)] · Medium

Q62. Verification based on a login ID and password is called:

Login ID और password के आधार पर जाँच कहलाती है:

(A) Logging in **(B)** Configuration
(C) Authentication **(D)** Confidentiality

[Source: RBI/IBPS computer-awareness banks (via ixambee)] · Medium

Q63. Access control based on a person's fingerprints is an example of:

किसी व्यक्ति की उंगलियों के निशान पर आधारित access control का उदाहरण है:

(A) Biometric identification **(B)** Characteristic identification
(C) Encryption **(D)** Logistics

[Source: SBI PO Computer Awareness PYQ (via bankersadda)] · Medium

Q64. Which one of the following is a key function of a firewall?

निम्न में से firewall का एक मुख्य काम कौन-सा है?

(A) Monitoring **(B)** Copying
(C) Deleting **(D)** Moving

[Source: RRB NTPC previous-year computer banks (via adda247 50-PYQ PDF)] · Medium

Q65. In IPSec tunnel mode, which parts of the original packet are encrypted?

IPSec tunnel mode में original packet के कौन-से भाग encrypt होते हैं?

(A) Only the header / केवल header **(B)** Only the payload / केवल payload
(C) The whole original packet / पूरा original packet **(D)** Nothing / कुछ नहीं

[Source: Practice — original (textbook-based)] · Medium

Q66. SSL/TLS works at which layer?

SSL/TLS किस layer पर काम करता है?

(A) Network **(B)** Transport (between transport and application) / Transport
(C) Physical **(D)** Data link

[Source: Practice — original (textbook-based)] · Medium

Q67. A VPN creates:

VPN बनाता है:

(A) A private tunnel over a public network / Public network पर private tunnel **(B)** A new physical cable / नई physical cable
(C) A virus filter / Virus filter **(D)** A new protocol / नया protocol

[Source: Practice — original (textbook-based)] · Medium

Q68. Which wireless security standard is the strongest among these?

इनमें से कौन-सी wireless security standard सबसे मज़बूत है?

(A) WEP **(B)** WPA
(C) WPA2 **(D)** Open

[Source: Practice — original (textbook-based)] · Medium

Q69. PGP (Pretty Good Privacy) is used for:

PGP प्रयोग होता है:

(A) Secure e-mail / सुरक्षित e-mail **(B)** Routing
(C) Cable testing **(D)** Address resolution

[Source: Practice — original (textbook-based)] · Medium

Q70. Which IPSec protocol provides encryption (confidentiality) of the packet data?

IPSec का कौन-सा protocol packet data की encryption (confidentiality) देता है?

(A) ESP **(B)** AH
(C) IKE **(D)** HMAC

[Source: Practice — original (textbook-based)] · Medium

Q71. In which IPSec mode is the original IP header encrypted together with the payload (as used in VPNs)?

IPSec की किस mode में मूल IP header भी payload के साथ encrypt होता है (जैसे VPN में)?

(A) Tunnel mode **(B)** Transport mode
(C) Transparent mode **(D)** Stealth mode

[Source: Practice — original (textbook-based)] · Medium

Q72. Which wireless security standard uses AES-CCMP encryption (the replacement for the weaker WEP/TKIP)?

कौन-सी wireless security standard AES-CCMP encryption प्रयोग करती है (कमज़ोर WEP/TKIP की जगह)?

(A) WPA2 **(B)** WEP
(C) WPA **(D)** Open authentication

[Source: Practice — original (SSC CGL/IBPS exam-pattern, textbook-based)] · Medium

Q73. What is Wired Equivalent Privacy (WEP) ?

(A) security algorithm for ethernet **(B)** security algorithm for wireless networks
(C) security algorithm for usb communication **(D)** none of the mentioned

[Source: Repo: DCN_MCQ-Ashwini-.pdf] · Medium

Q74. SYNC flooding attack belongs to a type of security attack known as _____

(A) sync flooding attack **(B)** active attack
(C) passive attack **(D)** denial-of-service attack

[Source: Repo: computer-networks [set 10].pdf] · Medium

Q75. IPsec defines two protocols: _____ and _____

(A) ah; ssl **(B)** pgp; esp
(C) ah; esp **(D)** pgp; ssl

[Source: Repo: computer-networks [set 10].pdf] · Medium

Q76. IP Security operates in which layer of the OSI model?

(A) network **(B)** transport
(C) application **(D)** physical

[Source: Repo: computer-networks [set 10].pdf] · Medium

Q77. In frame relay networks, extended address is used _____

(A) to increase the range of data link connection identifiers **(B)** for error detection
(C) for encryption **(D)** for error recovery

[Source: Repo: computer-networks [set 10].pdf] · Medium

Q78. In classless addressing, there are no classes but addresses are still granted in _____

(A) ips **(B)** blocks
(C) codes **(D)** sizes

[Source: Repo: computer-networks [set 11].pdf] · Medium

Q79. Network layer firewall works as a _____

(A) frame filter **(B)** packet filter
(C) signal filter **(D)** content filter

[Source: Repo: computer-networks [set 13].pdf] · Medium

Q80. Network layer firewall has two sub-categories called _____

(A) stateful firewall and stateless firewall **(B)** bit oriented firewall and byte oriented firewall
(C) frame firewall and packet firewall **(D)** network firewall and data firewall

[Source: Repo: computer-networks [set 13].pdf] · Medium

Q81. WPA2 is used for security in _____

(A) ethernet **(B)** bluetooth
(C) wi-fi **(D)** e-mail

[Source: Repo: computer-networks [set 13].pdf] · Medium

Q82. Network layer firewall has two sub-categories as _____

(A) state full firewall and stateless firewall **(B)** bit oriented firewall and byte oriented firewall
(C) frame firewall and packet firewall **(D)** network layer firewall and session layer firewall

[Source: Repo: computer-networks [set 13].pdf] · Medium

Q83. Which of the following is / are the types of firewall?

(A) packet filtering firewall **(B)** dual homed gateway firewall
(C) screen host firewall **(D)** dual host firewall

[Source: Repo: computer-networks [set 13].pdf] · Medium

Q84. What is one advantage of setting up a DMZ with two firewalls?

(A) you can control where traffic goes in three networks **(B)** you can do stateful packet filtering

(C) you can do load balancing **(D)** improved network performance

[Source: Repo: computer-networks [set 13].pdf] · Medium

Q85. A stateful firewall maintains a _____ which is a list of active connections

(A) routing table **(B)** bridging table

(C) state table **(D)** connection table

[Source: Repo: computer-networks [set 13].pdf] · Medium

Q86. _____ allows checking if a domain is available for registration

(A) domain check **(B)** domain dossier

(C) domain lookup **(D)** domain registers

[Source: Repo: computer-networks [set 13].pdf] · Medium

Q87. A point-to-point protocol over ethernet is a network protocol for _____

(A) encapsulating ppp frames inside ethernet frames

(B) encapsulating ehternet framse inside ppp frames

(C) for security of ethernet frames **(D)** for security of ppp frames

[Source: Repo: computer-networks [set 13].pdf] · Medium

Q88. In a piconet, there can be up to _____ parked nodes in the network

(A) 63 **(B)** 127

(C) 255 **(D)** 511

[Source: Repo: computer-networks [set 14].pdf] · Medium

Q89. What is A2DP (advanced audio distribution profile)?

(A) a bluetooth profile for streaming audio **(B)** a bluetooth profile for streaming video

(C) a bluetooth profile for security **(D)** a bluetooth profile for file management

[Source: Repo: computer-networks [set 14].pdf] · Medium

Q90. WiMAX provides _____

(A) simplex communication **(B)** half duplex communication

(C) full duplex communication **(D)** no communication

[Source: Repo: computer-networks [set 14].pdf] · Medium

Q91. RTP provides the facility of jitter _____

(A) media stream **(B)** expansion

(C) media modification **(D)** security

[Source: Repo: computer-networks [set 14].pdf] · Medium

Q92. When discussing IDS/IPS, what is a signature?

(A) an electronic signature used to authenticate the identity of a user on the network **(B)** attack-definition file

(C) it refers to “normal,” baseline network behavior

(D) it is used to authorize the users on a network

[Source: Repo: computer-networks [set 15].pdf] · Medium

Q93. A is a device that forwards packets between networks by processing the routing information included in the packet

(A) Bridge **(B)** Firewall

(C) Router **(D)** Hub

[Source: Repo: computer-networks [set 16].pdf] · Medium

Q94. What is a Firewall in Computer Network?

(A) The physical boundary of Network **(B)** An operating System of Computer Network

(C) A system designed to prevent unauthorized access

(D) A web browsing Software

[Source: Repo: computer-networks [set 17].pdf] · Medium

Q95. What is the Demilitarized Zone?

(A) The area between firewall & connection to an external network **(B)** The area between ISP to Military area

(C) The area surrounded by secured servers **(D)** The area surrounded by the Military

[Source: Repo: computer-networks [set 17].pdf] · Medium

Q96. Automatic repeat request error management mechanism is provided by

(A) logical link control sublayer **(B)** media access control sublayer

(C) network interface control sublayer **(D)** none of the mentioned

[Source: Repo: computer-networks [set 1].pdf] · Medium

Q97. What is autonegotiation?

(A) a procedure by which two connected devices choose common transmission parameters **(B)** a security algorithm

(C) a routing algorithm **(D)** none of the mentioned

[Source: Repo: computer-networks [set 3].pdf] · Medium

Q98. In asymmetric key cryptography, the private key is kept by

(A) sender **(B)** receiver
(C) sender and receiver **(D)** all the connected devices to the network

[Source: Repo: computer-networks [set 4].pdf] · Medium

Q99. Cryptanalysis is used

(A) to find some insecurity in a cryptographic scheme
(B) to increase the speed
(C) to encrypt the data **(D)** none of the mentioned

[Source: Repo: computer-networks [set 4].pdf] · Medium

Q100. Which one of the following is a cryptographic protocol used to secure HTTP connection?

(A) stream control transmission protocol (sctp) **(B)** transport layer security (tsl)
(C) explicit congestion notification (ecn) **(D)** resource reservation protocol

[Source: Repo: computer-networks [set 4].pdf] · Medium

Q101. The attackers a network of compromised devices known as

(A) internet **(B)** botnet
(C) telnet **(D)** d-net

[Source: Repo: computer-networks [set 4].pdf] · Medium

Q102. What is DHCP snooping?

(A) techniques applied to ensure the security of an existing dhcp infrastructure **(B)** encryption of the dhcp server requests
(C) algorithm for dhcp **(D)** none of the mentioned

[Source: Repo: computer-networks [set 4].pdf] · Medium

Q103. The password is sent to the server using _____ command

(A) passwd **(B)** pass
(C) password **(D)** none of the mentioned

[Source: Repo: computer-networks [set 5].pdf] · Medium

Q104. Presentation doesn’t perform _____ function

(A) translation **(B)** transmission
(C) encryption **(D)** compression

[Source: Repo: computer-networks [set 5].pdf] · Medium

Q105. SMTP connections secured by SSL are known as _____

(A) smtps **(B)** ssmtp
(C) snmp **(D)** starttls

[Source: Repo: computer-networks [set 8].pdf] · Medium

Q106. What is on-demand mail relay (ODMR)?

(A) protocol for smtp security **(B)** an smtp extension
(C) protocol for web pages **(D)** protocol for faster mail transfer

[Source: Repo: computer-networks [set 9].pdf] · Medium

Q107. SMTP defines _____

(A) message transport **(B)** message encryption
(C) message content **(D)** message password

[Source: Repo: computer-networks [set 9].pdf] · Medium

Q108. Which one of the following feature was present in SSH protocol, version 1?

(A) password changing **(B)** periodic replacement of session keys
(C) support for public-key certificates **(D)** none of the mentioned

[Source: Repo: computer-networks [set 9].pdf] · Medium

Q109. VPNs are financially speaking _____

(A) always more expensive than leased lines **(B)** always cheaper than leased lines
(C) usually cheaper than leased lines **(D)** usually more expensive than leased lines

[Source: Repo: computer-networks [set 9].pdf] · Medium

Q110. The main difference between SNMPv3 and SNMPv2 is _____

(A) management **(B)** integration
(C) classification **(D)** enhanced security

[Source: Repo: computer-networks [set 9].pdf] · Medium

Q111. Why do we require hamming codes?

(A) Error correction **(B)** Encryption only
(C) Decryption **(D)** Bit stuffing

[Source: Repo: Computer-Network-MCQ.pdf (new/ 2026 drop)] · Medium

Q112. Which of the communication modes support two way traffic but in only once direction of a time?

(A) Simplex **(B)** Half-duplex
(C) Three - quarter's duplex **(D)** Full duplex

[Source: Repo: networkmcq — GDC Ramban solved MCQ (new/ 2026 drop)] · Medium

Q113. A firewall is installed at the point where the secure internal network and untrusted external network meet which is also known as .....

(A) Chock point **(B)** meeting point
(C) firewall point **(D)** secure point

[Source: Repo: networkmcq — GDC Ramban solved MCQ (new/ 2026 drop)] · Medium

Q114. The .....is used to provide integrity check, authentication and encryption to IP datagram

(A) SSL **(B)** ESP
(C) TSL **(D)** PSL

[Source: Repo: networkmcq — GDC Ramban solved MCQ (new/ 2026 drop)] · Medium

Q115. The ..... Is a collection of protocols designed by Internet Engineering Task Force(IETF) to provide security for a packet at the Network level

(A) IPsec **(B)** Netsec
(C) Packetsec **(D)** Protocolsec

[Source: Repo: networkmcq — GDC Ramban solved MCQ (new/ 2026 drop)] · Medium

Q116. At the lower layer of SSL, a protocol for transferring data using a variety of predefined cipher and authentication combinations called the .....

(A) SSL handshake protocol **(B)** SSL authentication protocol
(C) SSL record protocol **(D)** SSL cipher protocol

[Source: Repo: networkmcq — GDC Ramban solved MCQ (new/ 2026 drop)] · Medium

Q117. On the upper layer of SSL, a protocol for initial authentication and transfer of encryption keys, called the .....

(A) SSL handshake protocol **(B)** SSL authentication protocol
(C) SSL record protocol GOVT.DEGREE COLLEGE RAMBAN(J&K) DEPARTMENT OF COMPUTER APPLICATION Solved Multiple Choice Questions on Computer networking **(D)** SSL cipher protocol

[Source: Repo: networkmcq — GDC Ramban solved MCQ (new/ 2026 drop)] · Medium

Q118. What is the advantage of setting up a DMZ (Demilitarized Zone) using a firewall?

Firewall का उपयोग करके DMZ (Demilitarized Zone) बनाने का क्या लाभ है?

(A) To create multiple zones of trust **(B)** Load balancing on the server
(C) Improving network performance **(D)** Bringing internet service into a home

[Source: RPSC Programmer 2024 Paper-1 (official key)] · Hard

Q119. Fragmentation has following demerits

(A) complicates routers **(B)** open to DOS attack
(C) overlapping of fragments **(D)** All of the mentioned

[Source: Repo: DCN_MCQ-Ashwini-.pdf] · Hard

Q120. The DoS attack is which the attacker establishes a large number of half-open or fully open TCP connections at the target host

(A) Vulnerability attack **(B)** Bandwidth flooding
(C) Connection flooding **(D)** All of the mentioned

[Source: Repo: DCN_MCQ-Ashwini-.pdf] · Hard

Q121. For encryption, WiMAX supports

(A) advanced encryption standard **(B)** triple data encryption standard
(C) both (a) and (b) **(D)** none of the mentioned

[Source: Repo: DCN_MCQ-Ashwini-.pdf] · Hard

Q122. ESP does not provide _____

(A) source authentication **(B)** data integrity
(C) privacy **(D)** error control

[Source: Repo: computer-networks [set 10].pdf] · Hard

Q123. Which of the following demerits does Fragmentation have?

(A) complicates routers **(B)** open to dos attack
(C) overlapping of fragments **(D)** all of the mentioned

[Source: Repo: computer-networks [set 11].pdf] · Hard

Q124. Which one of the following is not correct?

(A) rtcp provides canonical end-point identifiers to all session participants **(B)** rtcp reports are expected to be sent by all participants
(C) rtcp itself does not provide any flow encryption or authentication methods **(D)** rtcp handles the actual data delivery

[Source: Repo: computer-networks [set 15].pdf] · Hard

Q125. What is the purpose of a shadow honeypot?

(A) to flag attacks against known vulnerabilities **(B)** to help reduce false positives in a signature-based ids
(C) to randomly check suspicious traffic identified by an anomaly detection system **(D)** to enhance the accuracy of a traditional honeypot

[Source: Repo: computer-networks [set 15].pdf] · Hard

Q126. Which of the following is true of signature-based IDSes?

(A) they alert administrators to deviations from “normal” traffic behavior **(B)** they identify previously unknown attacks
(C) the technology is mature and reliable enough to use on production networks **(D)** they scan network traffic or packets to identify matches with attack-definition files

[Source: Repo: computer-networks [set 15].pdf] · Hard

Q127. Choose the correct statement from the following

(A) ppp can terminate the link at any time **(B)** ppp can terminate the link only during the link establishment phase
(C) ppp can terminate the link during the authentication phase **(D)** ppp can terminate the link during the callback control phase

[Source: Repo: computer-networks [set 15].pdf] · Hard

Q128. Which of the following is true for secure shell tunneling?

(A) To set up an SSH tunnel, one configures an SSH client to forward a specified local port **(B)** SSH tunnels provide a means to not bypassfirewalls
(C) All SSH clients support dynamic port forwarding
(D) Both A and B

[Source: Repo: computer-networks [set 16].pdf] · Hard

Q129. Which of the followings is not a direct implication of subnet masks in TCP/IP networks?

(A) Reduce network traffic **(B)** Increase network security
(C) Assignment of more IP addresses **(D)** Better network management

[Source: Repo: computer-networks [set 16].pdf] · Hard

Q130. The functionalities of the presentation layerinclude

(A) Data compression **(B)** Data encryption
(C) Data description **(D)** All of the mentioned

[Source: Repo: computer-networks [set 17].pdf] · Hard

Q131. Two devices are in network if

(A) a process in one device is able to exchange information with a process in another device **(B)** a process is running on both devices
(C) pids of the processes running of different devices are same **(D)** none of the mentioned

[Source: Repo: computer-networks [set 1].pdf] · Hard

Q132. In cryptography, what is cipher?

(A) algorithm for performing encryption and decryption **(B)** encrypted message
(C) both (a) and (b) **(D)** none of the mentioned

[Source: Repo: computer-networks [set 4].pdf] · Hard

Q133. Which one of the following algorithm is not used in asymmetric key cryptography?

(A) rsa algorithm **(B)** diffie-hellman algorithm
(C) electronic code book algorithm **(D)** none of the mentioned

[Source: Repo: computer-networks [set 4].pdf] · Hard

Q134. ElGamal encryption system is

(A) symmetric key encryption algorithm **(B)** asymmetric key encryption algorithm
(C) not an encryption algorithm **(D)** none of the mentioned

[Source: Repo: computer-networks [set 4].pdf] · Hard

Q135. Cryptographic hash function takes an arbitrary block of data and returns

(A) fixed size bit string **(B)** variable size bit string
(C) both (a) and (b) **(D)** none of the mentioned

[Source: Repo: computer-networks [set 4].pdf] · Hard

Q136. Which of the following is a form of DoS attack ?

(A) vulnerability attack **(B)** bandwidth flooding
(C) connection flooding **(D)** all of the mentioned

[Source: Repo: computer-networks [set 4].pdf] · Hard

Q137. The DoS attack is which the attacker sends deluge of packets to the targeted host

(A) vulnerability attack **(B)** bandwidth flooding
(C) connection flooding **(D)** all of the mentioned

[Source: Repo: computer-networks [set 4].pdf] · Hard

Q138. Firewalls are often configured to block

(A) udp traffic **(B)** tcp traffic
(C) both of the mentioned **(D)** none of the mentioned

[Source: Repo: computer-networks [set 4].pdf] · Hard

Q139. Transport services available to applications in one or another form _____

(A) reliable data transfer **(B)** timing
(C) security **(D)** all of the mentioned

[Source: Repo: computer-networks [set 8].pdf] · Hard

Q140. SSH uses _____ to authenticate the remote computer

(A) public-key cryptography **(B)** private-key cryptography
(C) any of public-key or private-key **(D)** both public-key & private-key

[Source: Repo: computer-networks [set 9].pdf] · Hard

Q141. Which one of the following authentication method is used by SSH?

(A) public-key **(B)** host based
(C) password **(D)** all of the mentioned

[Source: Repo: computer-networks [set 9].pdf] · Hard

Q142. Effective Storage management does not include _____

(A) security **(B)** backups
(C) reporting **(D)** connection

[Source: Repo: computer-networks [set 9].pdf] · Hard

Q143. Which of the following are the solutions to network security? i) Encryption ii) Authentication iii) Authorization iv) Non-repudiation

(A) i, ii and iii only **(B)** ii, iii and iv only
(C) i, iii and iv only **(D)** All i, ii, iii and iv 2 ..... is to protect data and passwords

[Source: Repo: networkmcq — GDC Ramban solved MCQ (new/ 2026 drop)] · Hard

Q144. The following protocols and system are commonly used to provide various degrees of security services in computer network. i) IP filtering ii) Reverse Address Translation iii) IP security Architecture (IPsec) iv) Firewalls v) Socks

(A) i, ii, iii and iv only **(B)** i, iii, iv and v only
(C) ii, iii, iv and v only **(D)** All i, ii, iii, iv and v

[Source: Repo: networkmcq — GDC Ramban solved MCQ (new/ 2026 drop)] · Hard

Q145. The components of IP security includes .....

(A) Authentication Header (AH) **(B)** Encapsulating Security Payload (ESP)
(C) Internet key Exchange (IKE) **(D)** All of the above 7 ..... is used to carry traffic of one protocol over network that does not support that protocol directly

[Source: Repo: networkmcq — GDC Ramban solved MCQ (new/ 2026 drop)] · Hard

Q146. In packet-filtering router, the following information can be external from the packet header. i) Source IP address ii) Destination IP address iii) TCP/UDP source port iv) ICMP message type v) TCP/UDP destination port

(A) i, ii, iii and iv only **(B)** i, iii, iv and v only
(C) ii, iii, iv and v only **(D)** All i, ii, iii, iv and v 10 ..... mode is used whenever either end of a security association is gateway

[Source: Repo: networkmcq — GDC Ramban solved MCQ (new/ 2026 drop)] · Hard

Answer Key — sirf letters (no solutions)

Q1: D **Q2:** C **Q3:** B **Q4:** C **Q5:** B **Q6:** B **Q7:** B **Q8:** A **Q9:** C **Q10:** A
Q11: B **Q12:** A **Q13:** A **Q14:** B **Q15:** A **Q16:** A **Q17:** A **Q18:** A **Q19:** A **Q20:** A
Q21: B **Q22:** D **Q23:** A **Q24:** B **Q25:** B **Q26:** C **Q27:** C **Q28:** C **Q29:** A **Q30:** B
Q31: A **Q32:** B **Q33:** C **Q34:** A **Q35:** B **Q36:** B **Q37:** B **Q38:** B **Q39:** B **Q40:** C
Q41: C **Q42:** B **Q43:** B **Q44:** B **Q45:** B **Q46:** A **Q47:** C **Q48:** C **Q49:** B **Q50:** C
Q51: A **Q52:** C **Q53:** C **Q54:** C **Q55:** A **Q56:** A **Q57:** C **Q58:** A **Q59:** A **Q60:** B
Q61: C **Q62:** C **Q63:** A **Q64:** A **Q65:** C **Q66:** B **Q67:** A **Q68:** C **Q69:** A **Q70:** A
Q71: A **Q72:** A **Q73:** B **Q74:** D **Q75:** C **Q76:** A **Q77:** A **Q78:** B **Q79:** B **Q80:** A
Q81: C **Q82:** A **Q83:** A **Q84:** C **Q85:** A **Q86:** A **Q87:** A **Q88:** C **Q89:** A **Q90:** C
Q91: A **Q92:** B **Q93:** C **Q94:** C **Q95:** A **Q96:** A **Q97:** A **Q98:** B **Q99:** A **Q100:** B
Q101: B **Q102:** A **Q103:** B **Q104:** B **Q105:** A **Q106:** B **Q107:** A **Q108:** D **Q109:** C **Q110:** D
Q111: A **Q112:** B **Q113:** A **Q114:** B **Q115:** A **Q116:** D **Q117:** A **Q118:** A **Q119:** D **Q120:** C
Q121: C **Q122:** D **Q123:** D **Q124:** D **Q125:** C **Q126:** D **Q127:** A **Q128:** A **Q129:** C **Q130:** D
Q131: A **Q132:** A **Q133:** C **Q134:** B **Q135:** A **Q136:** D **Q137:** B **Q138:** A **Q139:** D **Q140:** A
Q141: D **Q142:** D **Q143:** C **Q144:** A **Q145:** C **Q146:** B